

Doctor Handwritten Prescription Recognition Using Deep Learning and Transformer-Based Models

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Abstract—Handwritten medical prescriptions remain a significant source of medication errors due to illegible handwriting, ambiguous abbreviations, and diverse writing styles among physicians. Automated recognition of such prescriptions can substantially improve patient safety and healthcare efficiency. This project presents a deep learning-based framework for doctor handwritten prescription recognition using multiple state-of-the-art vision architectures. Convolutional neural network models, including EfficientNet-B3, DenseNet121, MobileNetV3, and ResNet50, are evaluated alongside transformer-based approaches such as Vision Transformer (ViT-B/16) and a hybrid CNN–Transformer architecture. The system processes scanned prescription images through preprocessing steps such as noise removal, normalization, and resizing before feature extraction and learning. A comparative analysis is conducted to assess recognition performance across models using standard evaluation metrics. Experimental results demonstrate that transformer-based and hybrid architectures exhibit improved robustness in handling complex handwriting patterns, while lightweight CNN models offer competitive accuracy with reduced computational cost. The findings highlight the trade-offs between performance and efficiency and underscore the potential of deep learning models in automating handwritten prescription recognition for real-world healthcare applications.

Index Terms—Handwritten Prescription Recognition, Deep Learning, Convolutional Neural Networks, Vision Transformer, Medical Image Analysis

I. INTRODUCTION

Handwritten medical prescriptions are still widely used in healthcare systems, particularly in developing regions, despite the availability of digital alternatives. However, poor handwriting, non-standard abbreviations, and variations in writing styles often lead to misinterpretation of prescriptions, increasing the risk of medication errors and compromising patient safety. According to multiple studies, illegible prescriptions remain a major contributor to dispensing errors, creating an urgent need for reliable automated recognition systems.

Traditional Optical Character Recognition (OCR) techniques struggle to accurately interpret handwritten medical text due to the cursive nature of writing, overlapping characters, and inconsistent spacing. Recent advances in deep learning have significantly improved handwritten text recognition by enabling models to learn complex visual features directly from data. Convolutional Neural Networks (CNNs) have demonstrated strong performance in extracting spatial features from images, while transformer-based architectures have shown

superior capability in modeling long-range dependencies and contextual relationships within visual data.

In the context of medical prescriptions, deep learning-based recognition systems must be both accurate and computationally efficient to be viable for real-world deployment. Lightweight CNN architectures such as MobileNetV3 offer faster inference with lower resource requirements, whereas deeper models like DenseNet121 and EfficientNet-B3 provide enhanced feature representation. Additionally, Vision Transformers (ViT) and hybrid CNN–Transformer models have emerged as powerful alternatives, capable of handling complex handwriting patterns that are difficult for purely convolutional models to recognize.

This project aims to design and evaluate a comprehensive handwritten prescription recognition framework using multiple state-of-the-art deep learning architectures, including EfficientNet-B3, DenseNet121, MobileNetV3, ResNet50, Vision Transformer (ViT-B/16), and a hybrid CNN + Transformer model. By conducting a comparative analysis of these models, the study seeks to identify performance trade-offs in terms of recognition accuracy, robustness, and computational efficiency. The outcomes of this work contribute to the ongoing effort to improve automated medical document analysis and support safer, more efficient healthcare systems.

II. LITERATURE REVIEW

Automated recognition of handwritten medical prescriptions has been an active research area due to its importance in reducing medication errors and improving healthcare delivery. Various studies have explored traditional machine learning, deep learning, and hybrid approaches to address the challenges posed by cursive handwriting, domain-specific terminology, and multilingual prescriptions.

Wijewardena [8] addressed the issue of illegible prescriptions by proposing an offline handwriting recognition system specifically tailored for medical prescriptions. The architecture combined a Convolutional Neural Network (CNN) for feature extraction with a Recurrent Neural Network (RNN) using Connectionist Temporal Classification (CTC) for sequence recognition. To enhance reliability, a domain-specific knowledge base of drug names was incorporated, allowing string-matching techniques to validate and correct recognized text.

Pavithiran et al. [3] focused on the challenges of multilingual handwritten prescriptions, where doctors often mix

English with regional languages. Their system utilized deep learning techniques in conjunction with Tesseract OCR to recognize handwritten text and included a translation module to convert the extracted information into the user's native language. This approach significantly improved accessibility and understanding for patients and pharmacists.

Ali et al. [1] introduced a transformer-based framework for extracting medicine names from handwritten prescriptions. Their methodology employed Mask R-CNN for accurate text segmentation followed by a Transformer-based Optical Character Recognition (TrOCR) model enhanced with multi-head attention and positional embeddings. The authors also contributed a novel dataset collected from diverse regions, demonstrating strong generalization across varying handwriting styles.

Maheswari et al. [2] explored the interpretation of doctors' handwritten notes using a combination of OCR techniques and deep learning. Their system applied CTC-based sequence modeling to organize and interpret recognized text, showing promising results in minimizing errors during patient data handling.

Shaw et al. [4] proposed a neural network-based prescription recognition system using a three-layer feedforward architecture trained on the EMNIST dataset. Their work emphasized character segmentation through grid-based scanning and achieved a reported training accuracy of 94.49%, highlighting the effectiveness of classical neural architectures for structured handwritten data.

Tabassum et al. [6] addressed online cursive handwriting recognition by introducing the Handwritten Medical Term Corpus containing English and Bangla medical terms. They proposed a novel data augmentation technique known as Stroke Rotation and Parallel-shift (SRP), which significantly improved recognition accuracy when combined with a Bidirectional LSTM model.

Yulianto [9] conducted a comparative study between conventional CNNs and the MobileNet architecture for prescription text recognition using a Bangladeshi dataset. The results showed that MobileNet outperformed standard CNN models, achieving 97% training accuracy and 88% testing accuracy, while maintaining computational efficiency suitable for web-based deployment.

Ullah et al. [7] proposed an AI-driven OCR framework integrating Transformer-based OCR (TrOCR) with Roboflow for image preprocessing. Their system achieved a Word Error Rate (WER) of 12.5% and a Character Error Rate (CER) of 8.7%, demonstrating strong potential for reducing medication errors through accurate digitization.

Shende et al. [5] evaluated multiple machine learning classifiers for prescription recognition, including Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Multi-Layer Perceptron (MLP). Their experiments showed that the MLP classifier achieved the highest accuracy of 97.5%, supported by Tesseract OCR and Natural Language Processing (NLP) techniques for structured data extraction.

Overall, existing research demonstrates that deep learning, particularly CNNs and transformer-based models, significantly outperforms traditional methods in handwritten prescription recognition. However, challenges such as handwriting variability, multilingual content, and computational constraints remain, motivating further comparative studies across diverse architectures, as undertaken in this project.

III. METHODOLOGY

This section describes the overall framework used for doctor handwritten prescription recognition, including the dataset characteristics, preprocessing techniques, and the learning strategies adopted for training multiple deep learning models.

A. Data Description

The dataset used in this project consists of scanned images of handwritten medical prescriptions. These images exhibit significant variability in handwriting styles, stroke thickness, spacing, and layout, reflecting real-world clinical conditions. The prescriptions contain textual information such as medicine names, dosage instructions, and abbreviations commonly used by physicians.

All images were labeled according to their corresponding textual content and organized into training and validation sets. The dataset was selected to ensure diversity in handwriting patterns and image quality, enabling robust evaluation of different deep learning architectures under realistic conditions.

B. Data Preprocessing

Effective preprocessing is essential for improving recognition accuracy in handwritten text analysis. Initially, all prescription images were resized to a fixed resolution compatible with the input requirements of the selected models. Color images were converted to grayscale where applicable to reduce computational complexity while preserving essential structural information.

Noise removal techniques, including Gaussian filtering and contrast normalization, were applied to minimize background artifacts and scanning irregularities. Image normalization was performed to standardize pixel intensity values, facilitating stable and efficient training. Data augmentation techniques such as rotation, scaling, and horizontal translation were employed to increase dataset diversity and enhance model generalization, particularly for cursive and irregular handwriting.

C. Learning Phase

During the learning phase, multiple deep learning architectures were trained and evaluated for handwritten prescription recognition. Convolutional neural network models, including EfficientNet-B3, DenseNet121, MobileNetV3, and ResNet50, were used to extract hierarchical spatial features from prescription images. These models leverage convolutional layers to capture local patterns such as strokes, curves, and character shapes.

In addition to CNN-based approaches, transformer-based architectures were explored to capture global contextual relationships within images. The Vision Transformer (ViT-B/16)

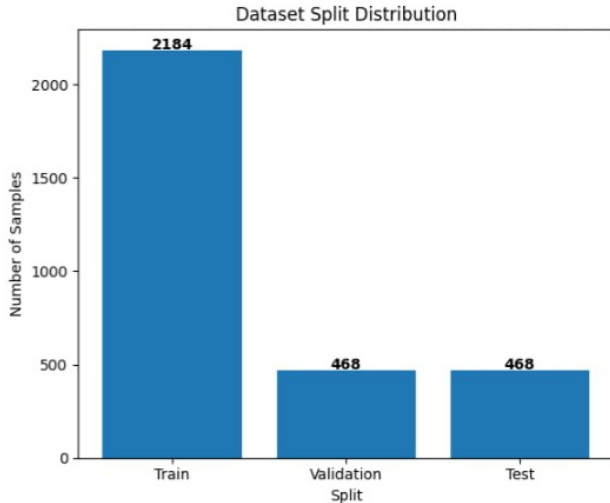


Fig. 1. Distribution of dataset splits used for training, validation, and testing.

model processes images as sequences of patches and employs self-attention mechanisms to model long-range dependencies. A hybrid CNN + Transformer architecture was also implemented, combining convolutional feature extraction with transformer-based sequence modeling to leverage the strengths of both paradigms.

All models were trained using supervised learning with labeled data. Optimization was performed using stochastic gradient-based methods, and categorical cross-entropy loss was used as the objective function. Model performance was monitored using validation data to prevent overfitting, and early stopping techniques were applied where necessary.

IV. RESULTS AND DISCUSSION

This section presents the experimental results obtained from evaluating multiple deep learning models for handwritten prescription recognition and provides a detailed discussion of their performance and learned representations.

A. Performance Metrics

To quantitatively evaluate the performance of the proposed models, standard classification and recognition metrics were employed. Accuracy was used as the primary metric to measure the overall correctness of predictions. Precision, recall, and F1-score were also computed to assess the balance between false positives and false negatives, which is particularly important in medical applications where misclassification can lead to serious consequences.

During training and validation, loss curves were monitored to analyze convergence behavior and detect overfitting. Models were compared under identical training conditions to ensure a fair evaluation. Transformer-based models additionally benefited from attention-based mechanisms, which contributed to improved recognition of complex and ambiguous handwritten patterns.

B. Feature Importance

Feature importance was analyzed to understand how different models learned discriminative representations from handwritten prescription images. CNN-based architectures such as ResNet50 and DenseNet121 demonstrated strong sensitivity to local visual patterns, including character strokes, edges, and curvature, which are essential for recognizing individual handwritten characters.

EfficientNet-B3 achieved a balanced representation by scaling depth, width, and resolution, enabling efficient extraction of both low-level and high-level features. MobileNetV3, while lightweight, focused on essential structural components of handwritten text, making it suitable for resource-constrained environments.

Transformer-based models, particularly ViT-B/16 and the hybrid CNN + Transformer architecture, leveraged self-attention mechanisms to capture long-range dependencies and contextual relationships across image regions. This capability allowed the models to better handle irregular spacing, overlapping characters, and stylistic variations in handwriting.

C. Comparative Model Analysis

A comparative analysis revealed clear trade-offs between model complexity, accuracy, and computational efficiency. Vision Transformer and hybrid CNN + Transformer models consistently showed superior robustness in recognizing complex handwritten prescriptions, especially in cases involving cursive writing and overlapping text. However, these models required higher computational resources and longer training times.

Among the CNN-based models, DenseNet121 and EfficientNet-B3 achieved strong recognition performance due to their effective feature reuse and compound scaling strategies. MobileNetV3 offered the fastest inference time with competitive accuracy, making it a practical choice for real-time or deployment-oriented applications.

Overall, the results indicate that while transformer-based architectures provide higher accuracy and robustness, lightweight CNN models remain valuable for scenarios where efficiency and speed are critical. The findings underscore the importance of selecting model architectures based on application-specific requirements.

V. CONCLUSION

This project presented a comprehensive deep learning-based framework for recognizing doctor handwritten medical prescriptions. Multiple state-of-the-art architectures, including EfficientNet-B3, DenseNet121, MobileNetV3, ResNet50, Vision Transformer (ViT-B/16), and a hybrid CNN + Transformer model, were implemented and evaluated to analyze their effectiveness in handling complex handwritten medical text.

Experimental results demonstrated that transformer-based and hybrid architectures provided improved robustness in recognizing cursive handwriting and irregular prescription layouts due to their ability to model global contextual relationships.

At the same time, lightweight convolutional models such as MobileNetV3 achieved competitive accuracy with significantly lower computational overhead, making them suitable for real-time and resource-constrained applications. The comparative analysis highlighted important trade-offs between recognition accuracy, model complexity, and efficiency.

Overall, the findings indicate that deep learning approaches have strong potential to reduce medication errors caused by illegible prescriptions and to support safer healthcare workflows. Future work may include expanding the dataset to cover a wider range of handwriting styles and languages, integrating domain-specific medical knowledge for post-processing, and deploying the system as a real-time clinical decision support tool.

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